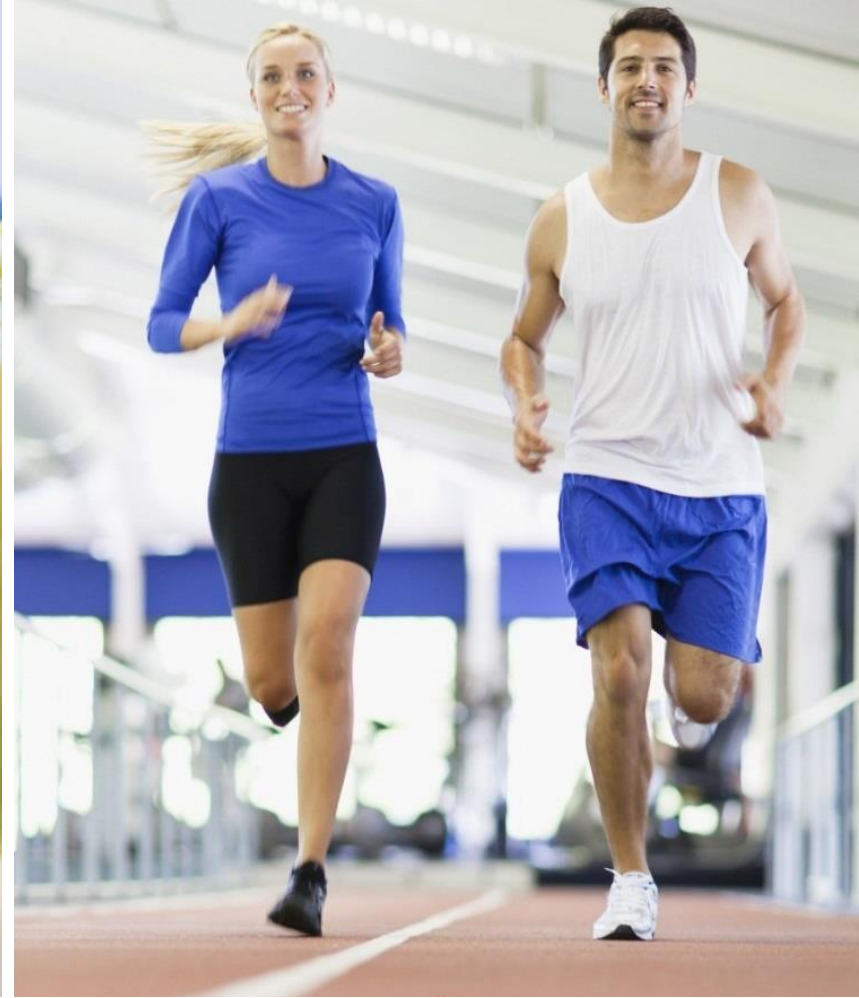




HEALTHCARE DATA MODELING

CAPSTONE PROJECT PRESENTATION
BY ARINDAM SAHA



BUSINESS PROBLEM UNDERSTANDING - 1

BUSINESS PROBLEM UNDERSTANDING

Business Problem:

- Healthcare costs can place a **significant financial burden** on individuals without adequate insurance coverage.
- Medical insurance companies face the challenge of optimizing **insurance premiums** while managing risk.
- Our goal is to develop a data-driven model that accurately predicts the optimum insurance cost for individuals, ensuring fair premiums and sustainable business operations.

BUSINESS PROBLEM UNDERSTANDING

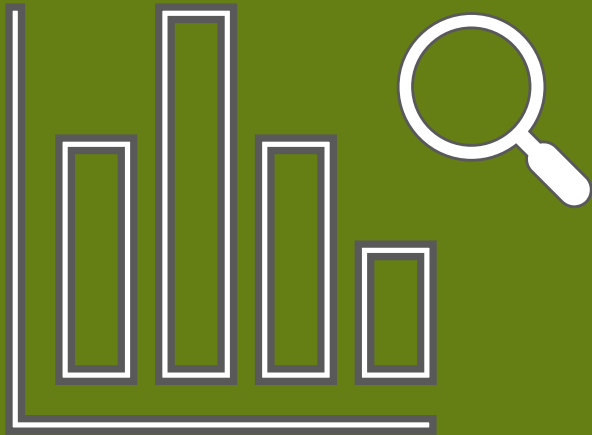
Constrains:

- Limited availability of real-time health data for accurate predictions.
- Privacy and compliance regulations restrict data usage.
- Model accuracy may vary depending on data quality and missing values.
- Potential bias in historical data impacting prediction fairness.

BUSINESS PROBLEM UNDERSTANDING

Objectives:

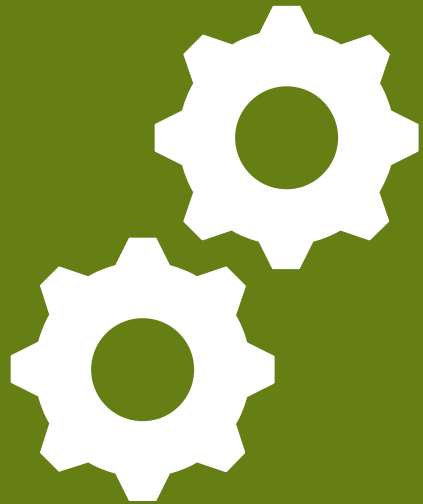
- Develop a **machine learning model** to estimate insurance premiums accurately.
- Identify key independent factors impacting insurance costs.
- Support insurers in offering fair, **personalized premium rates**.
- Empower individuals with insights to manage their health proactively.



EXPLORATORY DATA ANALYSIS (EDA) - 2

EXPLORATORY DATA ANALYSIS (EDA)

- The data had in total 24 columns and 25000 rows.
- The dataset consists of 8 categorical columns and 16 numerical columns.
- The BMI column had 990 missing values, while the Year_last_admitted column contained 11,881 missing values in the dataset.
- As the number of years since last admission increases, the insurance costs also increased.
- High insurance cost has direct relation with the overweight.
- Weight change in last one year had a significant negative correlation in determining the insurance cost.



FEATURE ENGINEERING - 3

FEATURE ENGINEERING

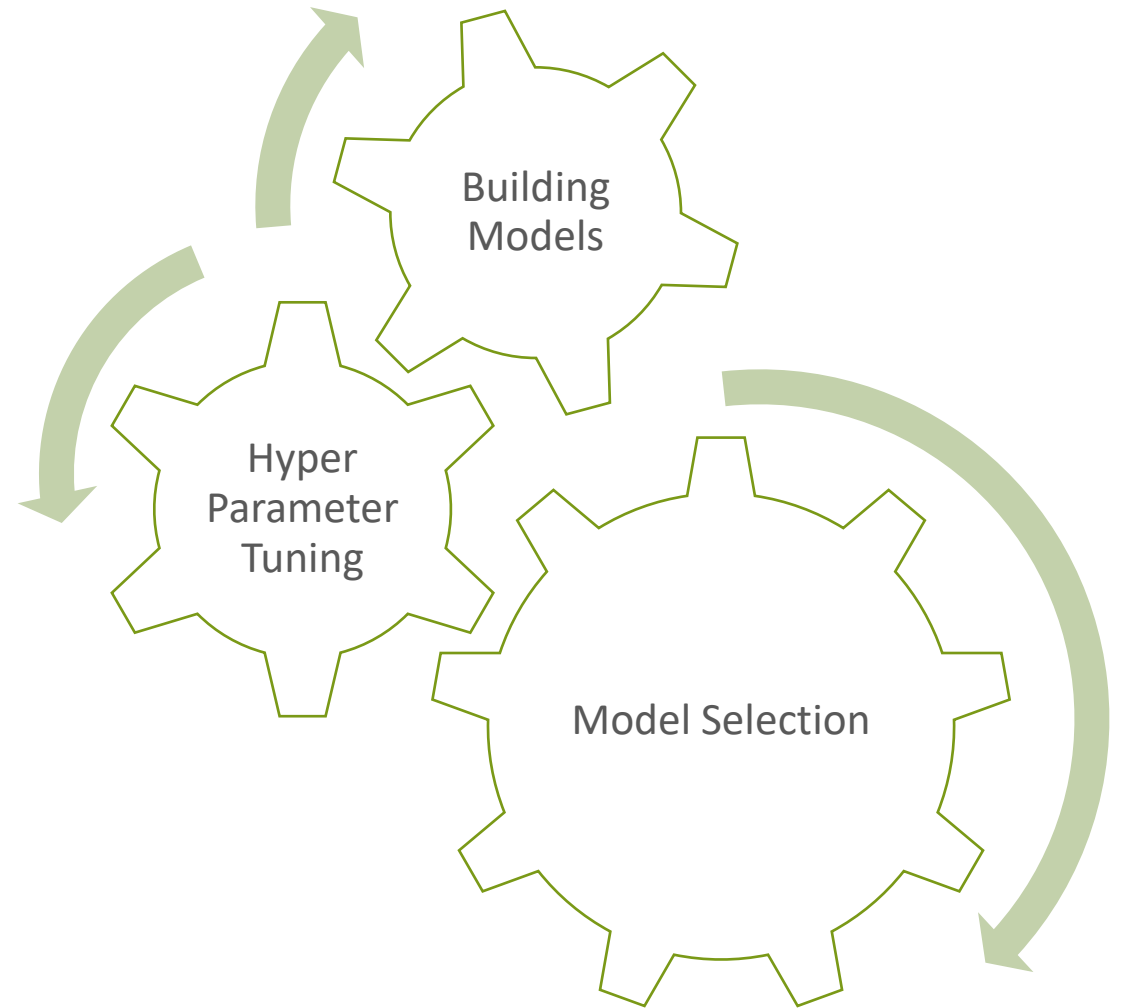
- Dropped column applicant_id which is not adding any information.
- A new column (Years_since_last_admission) has been added to indicate the number of years passed since the last admission ($\text{Years_since_last_admission} = \text{current year} - \text{Year_last_admitted}$).
- Missing values in the BMI column were filled with the mean BMI based on gender.
- Missing values in Years_since_last_admission were imputed with -1.
- For data modelling we have not considered location field as it has no impact on the insurance cost.
- The following columns were dropped: 'Location', and 'Year_last_admitted' as they don't contribute any relation with the dependent variable.
- In the given dataset, categorical variables were converted into numerical values to prepare the data for regression modelling.



MODELLING - 4

MODELLING

- Splitting of the data in 70:30 ratio for train and test set was done.
- Number of rows in train data = 17500
- Number of rows in test data = 7500
- Models analyzed : Logistic Regression, Decision Tree, Bagging, Boosting, Random Forest.



MODEL COMPARISON

TRAIN RESULTS

Models	R ²	Adj. R ²	RMSE	MAE	MAPE(%)
Linear Regression train	0.8803	0.8803	0.23	0.18	1.82
Dtree train	1	1	0	0	0
Tuned Dtree train	0.7093	0.7089	0.36	0.29	2.91
Bagging train	0.99	0.99	0.07	0.04	0.46
Tuned Bagging train	0.9912	0.9912	0.06	0.05	0.47
RF train	0.9926	0.9926	0.06	0.04	0.43
Tuned RF train	0.9869	0.9869	0.08	0.06	0.57
Ada Boost train	0.9333	0.9332	0.17	0.13	1.33
Tuned Ada Boost train	0.9323	0.9322	0.17	0.13	1.34
Gradient Boost train	0.95	0.9499	0.15	0.11	1.12
Tuned Gradient Boost train	0.9557	0.9557	0.14	0.1	1.06
XGBoost train	0.978	0.978	0.1	0.07	0.75
Tuned XGBoost train	0.9494	0.9493	0.15	0.11	1.14

TEST RESULTS

Models	R ²	Adj. R ²	RMSE	MAE	MAPE(%)
Linear Regression test	0.8832	0.883	0.22	0.17	1.78
DTree test	0.8923	0.892	0.21	0.15	1.57
Tuned DTree test	0.6966	0.6957	0.36	0.29	2.93
Bagging test	0.9404	0.9403	0.16	0.12	1.21
Tuned Bagging test	0.9419	0.9417	0.16	0.12	1.21
RF test	0.9459	0.9458	0.15	0.11	1.15
Tuned RF test	0.9474	0.9472	0.15	0.11	1.14
Ada Boost test	0.9333	0.9331	0.17	0.13	1.32
Tuned Ada Boost test	0.9322	0.932	0.17	0.13	1.33
Gradient Boost test	0.9483	0.9481	0.15	0.11	1.14
Tuned Gradient Boost test	0.949	0.9489	0.15	0.11	1.13
XGBoost test	0.9446	0.9445	0.15	0.11	1.17
Tuned XGBoost test	0.9486	0.9485	0.15	0.11	1.14

Remarks

Low Performance

Overfitting Model

Overfitting Model with Improved Accuracy

Performance Difference (Reduced Overfitting)

Performance Difference with Better Generalization

Performance Difference (Good Balance)

Optimum Model

Low Performance with High Bias

Low Performance with Slight Improvement

Performance Difference (Good but Slightly Overfitting)

Optimum Model with Better Performance

Performance Difference (High Accuracy, Low Overfitting)

Optimum Model with Strong Generalization

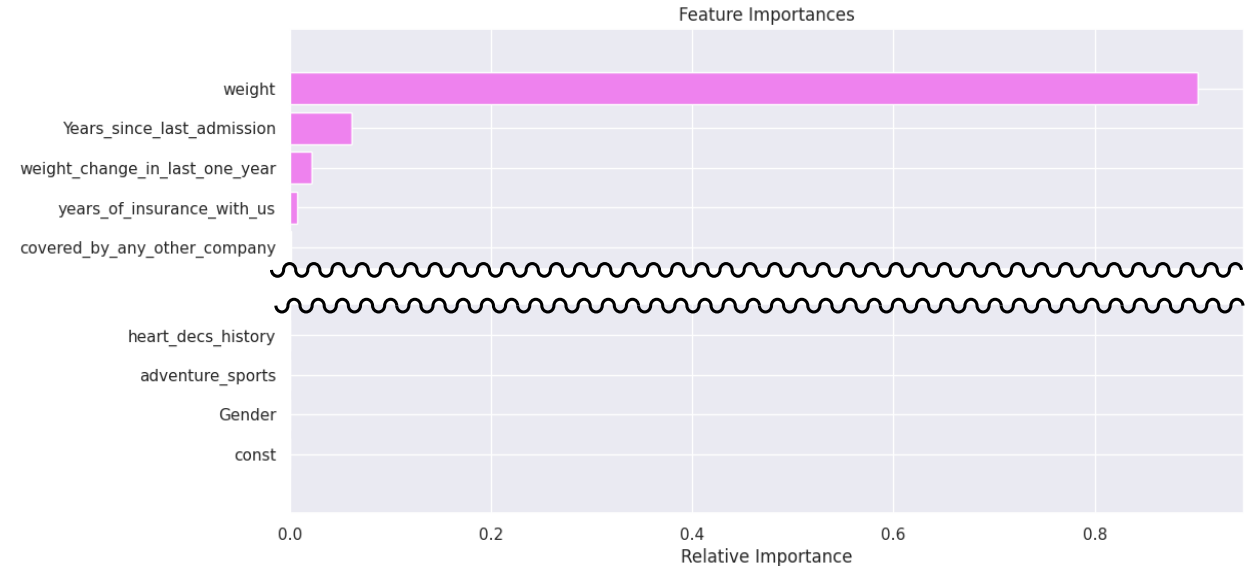
MODEL SELECTION

Dominant Feature:

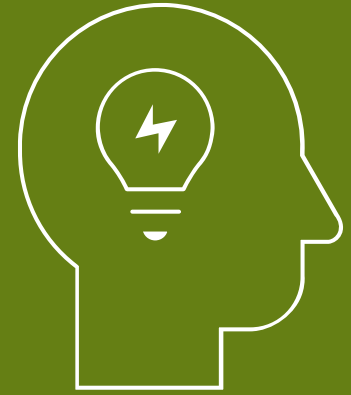
- **Weight** is the most influential feature with an importance score of **0.902**, indicating it has a significant impact on model predictions. This heavy reliance suggests that weight alone contributes disproportionately to the outcome.

Moderately Important Features:

- **Years Since Last Admission** (0.061) and **Weight Change in Last One Year** (0.021) are the next most impactful variables, indicating some influence on the target variable.



TUNED GRADIENT
BOOSTING



1. Offer Incentives for a Healthier Lifestyle

Discounts to physically active customers, smoking cessation programs, free annual health checkup.

2. Personalized & Dynamic Pricing Models

AI-driven risk scoring, introduce wearable devices, offer customized plans.

3. Expand Preventive Healthcare Programs

Partner with gyms, dieticians, and mental health programs, promote telemedicine services, offer discounts on health screening.

4. Fraud Prevention & Risk Reduction

Monitor sudden weight gain/loss, detect patterns with ML.

INSIGHTS AND
BUSINESS
RECOMMENDATIONS - 5

Thank You



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